INCH-POUND

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MILITARY ASSOCIATED DETAIL SPECIFICATION

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR ASSEMBLY, MK 113 MOD O

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Mk 113 Mod 0 Rocket Motor Assembly described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification covers the requirements unique to the Rocket Motor Assembly Mk 113 Mod 0 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the basic specification, MIL-A-85097, and form a part of this specification to the extent specified herein.

SPECIFICATIONS

NAVAL SURFACE WARFARE CENTER (CAGE Code 14083)

511-174TD-004 Propellant, NOSIH-BC-11

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL673AS200 Rocket Motor Assembly, Mk 113 Mod 0
DL673AS205 WORD Rocket Motor Assembly

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Code 8420, 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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*	673AS200	Rocket Motor Assembly, Mk 113 Mod 0
*	673AS205	WORD Rocket Motor Assembly

(Application for copies of specifications and drawings should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

OP 2165	Navy Transportation Safety Handbook for Hazardous Materials

(Application for copies should be addressed to the Aviation Supply Office, Naval Publications and Forms Directorate, Attn: 1053, Building 26, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

3. REQUIREMENTS

3.1 General. The requirements in this associated detail specification, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Rocket Motor Assembly Mk 113 Mod 0 shall conform to DL673AS200 and the Mk 113 Wind Oriented Rocket Deployment (WORD) Motor Assembly shall conform to DL673AS205.

3.2 Item description. The Rocket Motor Assembly Mk 113 Mod 0 specified herein is comprised of a WORD rocket motor mated to a drogue release mechanism.

3.3 Preproduction.

3.3.1 Propellant aging. When tested in accordance with MIL-A-85097, the WORD motor propellant shall meet the aging requirements specified in 511-174TD-004.

3.4 Preproduction/production.

3.4.1 WORD motor propellant. The propellant shall conform to 511-174TD-004. The contractor shall submit all test results in accordance with 511-174TD-004 and MIL-A-85097.

3.4.2 WORD motor ballistic performance. The ballistic performance shall conform to Table I.

3.4.3 Drogue release assembly performance. Upon application of 1,000 psig to either inlet port (see 4.1.3), the release shall actuate within 400 to 1,000 psig and shall in no way hinder deployment of the WORD rocket motor in any configuration/load arrangement.

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* TABLE I. Ballistic performance requirements.

Characteristic	Value		Temperature (± 5°F)
	Min	Max	
Impulse over 390 ms (lb-s) (see 6.2.1)	47	--	-40, 160
Ignition delay (ms)	-	10	-40, 160
Actuation force (lb)	15	25	-40, 160
Rise time (ms) (see 6.2.2)	-	30	-40, 160
Thrust (lb)	-	270	-40, 160

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction/production.

4.1.1 Conditioning period. Time for temperature conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: not less than 3 hours.
b. Temperature cycling: not less than 4 hours.

4.1.2 Radiographic inspection. Units shall be radiographed in two views 90° apart, as specified by Drawing 673AS200 (Mk 113 Mod O WORD/Drogue Assembly) or by Drawing 673AS205 (WORD Motor Assembly), as appropriate.

* **4.1.3 Drogue release testing.** Drogue release testing shall be performed on the drogue release assemblies in accordance with the following:

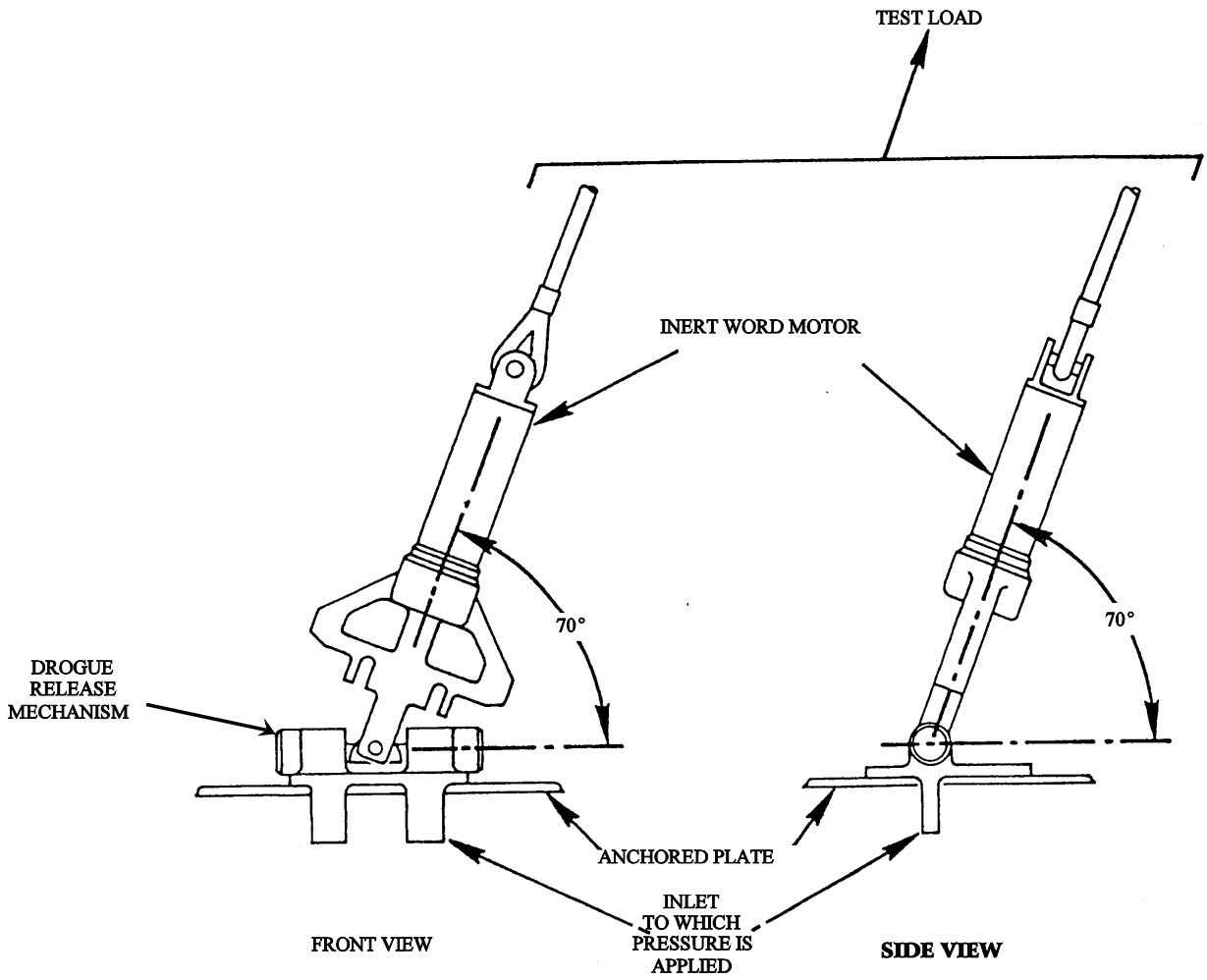
* a. Half of the test units shall be conditioned at -40°F and the other half at 160°F for the time specified in 4.1.1a.

* b. There shall be a light test load of 25 to 35 lb, and a heavy test load of 1000 to 1025 lb. The light and heavy test loads shall be distributed evenly over the two conditioning temperatures.

* c. Actuation pressure shall be regulated to 1,000 to 1,025 psig and pressurization rate shall be regulated to 30,000 to 50,000 psig/sec.

* d. Only one side/port of each drogue release shall be tested. Either side may be tested provided the load arrangement in Figure 1 is maintained.

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FIGURE 1. Drogue release test configuration.

5. PACKAGING

5.1 Packaging. All packaging requirements shall be in accordance with MIL-A-85097 and the applicable contract.

5.2 Marking. The shipping containers shall be marked "Explosive Power Device, Class B - Keep Fire Away" in accordance with OP 2165. See the applicable contract and the latest revision of MIL-A-85097 for additional marking requirements.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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* **6.1 Intended use.** The Mk 113 Mod 0 Rocket Motor Assembly assists in the deployment of the main parachute in T/AV-8B, A-7E, and TA-7C aircraft.

6.2 Definitions. For the purpose of this specification, the following definitions apply:

* **6.2.1 Impulse over 390 ms.** The total impulse over the interval from the start of the burn time (time at 10% of the maximum thrust during the initial rise on the thrust curve t_{b0}) to 390 ms of burn time (t_{b0} + 390 ms) or the end of the burn time, whichever occurs first.

6.2.2 Rise time. The interval from 10% to 75% of the maximum thrust during the initial rise on the thrust curve occurring just after ignition.

* **6.3 Changes from previous issue.** The margins of this revision are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous revision were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous revision.

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-85097/11 C	2. DOCUMENT DATE (YYMMDD) 24 October 1994

3. DOCUMENT TITLE
AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR ASSEMBLY, MK 113 MOD 0

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700 (2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Standardization Team (Code 840M) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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